Omitted Variables and Conditional Independence

Lecture 7 (R code)

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Reading

This lecture covers material from Wooldridge (2025):

- Chapter 3.3 "The Expected Value of the OLS Estimators"
- Chapter 3.7 "Several Scenarios for Applying Multiple Regression"
- Chapter 3.8 "Including"Bad Controls" in Regression Analysis"
- Chapter 4.7 "Revisiting Causal Effects and Policy Analysis"
- Chapter 7.6 "More on Policy Analysis and Program Evaluation"
  - Chapter 7.6a "Program Evaluation and Unrestricted Regression Adjustment"

- Chapter 9.1 “Function form Misspecification”
  - Chapter 9.1a “RESET as a Test for Functional Form Misspecification”

We will leave out 9.1b “Tests against Nonnested Alternatives”

Example: Selective Education

> **Note**
>
> The following example comes from Angrist and Pischke (2014, ch. 2) *Mastering Metrics*. The book is an excellent read and worth getting a copy of. The example is inspired by a famous paper *Estimating the payoff to attending a more selective college: An application of selection on observables and unobservables* (Dale and Krueger 2002). Another excellent read.

Self-selection

Suppose you want to estimate the (causal) impact of attending a selective (p-rivate) university on future earnings (y)[1]

- Individuals *self-select* into private education
  - $\Rightarrow p$ is *endogenous* to factors that plausibly determine future earnings.

> **Terminology**
>
> Take note of how we use these words: self-select, endogenous, exogenous, uncorrelated, independent, unconfounded, etc. They all have important, but slightly different meanings.

Confounding factors

Given a simple (s-hort) linear regression,[2]

$$y_i = \alpha^s + \beta^s p_i + v_i$$

[1] In Dale and Krueger (2002) compare Ivy League graduates against State university ones

[2] I am borrowing the notation from Angrist and Pischke (2014), who discusses these as the long and short model. So, the superscript ^{s} is for short. Wooldridge does not use this notation, but it is helpful to keep track of the different parameters.

there are likely many **confounding factors** (parental education, ability, etc.) that determine both y and p.

Control variables

You might control for these confounding factors by estimating a multivariate model

$$y_i = \alpha + \beta p + \mathbf{x}_i\gamma + u_i$$

Traditional Story: Omitted Variables

'True' model

The 'true' model is given by,[3]

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$$

with MLR 1-4 holding.

- $E[\hat{\beta}_1] = \beta_1$
- $\hat{\beta}_1 \to_p \beta_1$ (even under MLR 4a)

Omitted variable

In place of the 'true' model, you estimate the 'short' model,

$$y_i = \beta_0^s + \beta_1^s x_{i1} + v_i$$

which **omits** x_2.

> **Model Misspecification**
>
> Wooldridge (2025, 85) discusses this problem as an example of misspecification analysis. The 'short' model is misspecified since it excludes variables that belong in the 'true'

[3] You can include more than 2 regressors. It just makes the math a little more tedious.

model.

What then is the relationship between,

$$\beta_1^s \stackrel{?}{=} \beta_1$$

And, if you estimate the short model, what - if anything - do you learn about the parameters from the longer ('true') model?

Omitted variable formula

From Lecture 1, we know that β_1^s can be defined as,[4]

$$\beta_1^s = \frac{Cov(x_{i1}, y_i)}{Var(x_{i1})}$$

If we plug in y_i from the 'true' model, then

$$\begin{aligned}
\beta_1^s =& \frac{Cov(x_{i1}, y_i)}{Var(x_{i1})} \\
=& \frac{Cov(x_{i1}, \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i)}{Var(x_{i1})} \\
=& \underbrace{\frac{Cov(x_{i1}, \beta_0)}{Var(x_{i1})}}_{=0} + \underbrace{\frac{Cov(x_{i1}, \beta_1 x_{i1})}{Var(x_{i1})}}_{=\beta_1} + \frac{Cov(x_{i1}, \beta_2 x_{i2})}{Var(x_{i1})} + \underbrace{\frac{Cov(x_{i1}, u_i)}{Var(x_{i1})}}_{=0} \\
=& \beta_1 + \beta_2 \frac{Cov(x_{i1}, x_{i2})}{Var(x_{i1})}
\end{aligned}$$

[4] To define the population parameter in this way, we require MLR 1-4(a).

OV formula

So,

$$\beta_1^s = \beta_1 + \beta_2 \delta_1$$

where,

- β_2 is the parameter on x_2 in the 'true' model; i.e. the impact of x_2 on y.
- $\delta_1 = \frac{Cov(x_{i1}, x_{i2})}{Var(x_{i1})}$ is the slope parameter from a simple regression,

$$x_{i2} = \delta_0 + \delta_1 x_{i1} + \xi_i$$

  i.e., the impact of x_1 on x_2.

OV 'bias'

Consider the (conditional) expectation of the OLS estimator of the short regression,[5]

$$E[\hat{\beta}_1^s | X] = \beta_1 + \underbrace{\beta_2 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}}_{\text{'bias'}}$$

and[6]

$$p\lim \ \hat{\beta}_1^s = \beta_1 + \beta_2 \underbrace{\frac{Cov(x_{i1}, x_{i2})}{Var(x_{i1})}}_{\delta_1}$$

> **Why 'bias'?**
>
> Why do we call this a bias? An estimator of is biased if $E[\hat{\beta}_j] \neq \beta_j$. We call this a bias because of the model misspecification framing. The question here is whether the misspecified model (the 'short' model) can be used to estimate the 'true' model.
> Consider the long model,
>
> $$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$$

[5] This formula can be found on p. 87 of Wooldridge (2025). Wooldridge uses $\tilde{\beta}_1$ in place of $\hat{\beta}_1^s$.
[6] This formula can be found on p. 171 of Wooldridge (2025).

If you propose a short model with the same population parameters,

$$y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i$$

then $\varepsilon_i = \beta_2 x_{i2} + u_i$. Well, suppose you assert that β_1 denotes the causal impact of x_1 on y. Then, the short model essentially asks whether you can estimate this causal relationship using a model that omits x_2.
Since, the parameters are the same,

$$Cov(x_{i1}, \varepsilon_i) = \beta_2 Cov(x_{i1}, x_{i2}) + Cov(x_{i1}, u_i)$$

Both MLR 4 and 4a fail if x_1 and x_2 are correlated OR $\beta_2 = 0$. Implying that the OLS estimator for is otherwise biased (for β_1).
However, you could argue that $\hat{\beta}_1^s$ is unbiased estimator of β_1^s. And the real difference here is that β_1^s need not equal β_1. So, the discussion actually concerns the relationship between the population parameters and not the biasedness of the estimator. This second approach better fits the modern causal identification discussion which distinguishes well identified population moments (e.g. mean difference between two groups) and causal parameters/estimands (e.g. ATE).

> "This important formula is often motivated by the notion that a longer regression – one with controls–has a causal interpretation, while a shorter rgression does not. The coefficients on the variables included in the shorter regression are therefroe said to be biased. In fact, the OVB formula is a mechanical link between coefficient vectors that applies to short and long regressions whether or not the longer regression is causal." (Angrist and Pischke 2009, 59)

Sample analogue

The same formula holds for the OLS estimator:

$$\hat{\beta}_1^s = \hat{\beta}_1 + \hat{\beta}_2 \hat{\delta}_1$$

- The fact that $\hat{\delta}_1$ is from a regression of x_2 on x_1 relates to the Frisch-Waugh Theorem of Lecture 2.
- The multivariate estimate is a simple estimate *minus* the estimated OVB term

$$\hat{\beta}_1 = \hat{\beta}_1^s - \hat{\beta}_2 \hat{\delta}_1$$

  - But this formula works for more general case of adding additional regressors.

Demonstration

```
use "$data_dir\morg24_bothgen.dta", clear

qui reg lnwage edu
  est store mod_short

qui reg lnwage edu exp
  est store mod_long
  scalar beta1 = _b[edu]
  scalar beta2 = _b[exp]

qui reg exp edu
  est store mod_x2x1
  scalar delta1 = _b[edu]

etable, est(mod_short mod_long mod_x2x1) column(depvar) eqrecode(lnwage = xb exp = xb) novarl
```

```
                        lnwage   lnwage     exp
-------------------------------------------------
edu                      0.1021   0.1074  -0.8341
                        (0.0027) (0.0027) (0.0694)
exp                               0.0063
                                 (0.0005)
_cons                    1.9324   1.7119  34.7376
                        (0.0395) (0.0433) (1.0154)
Number of observations     5000     5000     5000
-------------------------------------------------
```

```
dis beta1 + beta2*delta1
```

```
.10206761
```

OVB sign

$$OVB = \beta_1^s - \beta_1 = \beta_2 \delta_1$$

Table 1: Direction of OVB

	$\delta_1 < 0$	$\delta_1 = 0$	$\delta_1 > 0$
$\beta_2 < 0$	Positive	Zero	Negative ('downward')
$\beta_2 = 0$	Zero	Zero	Zero
$\beta_2 > 0$	Negative	Zero	Positive ('upward')

- You can predict how the exclusion of a potential confounding factor may 'bias' your estimates.
- Alternatively, you can predict how the inclusion of a regressor will (in expectation) alter the existing parameters in a model.

Example: Selective Education

If you consider the single control $x_i = SAT_i$

$$y_i = \alpha^s + \beta^s p_i + v_i$$

versus

$$y_i = \alpha + \beta p_i + \gamma SAT_i + u_i$$

How do you expect β to change, relative to β^s?

Limitations

1. We can only "control" for factors which we observe in the data
    - Caveat:
        - researchers like Dale and Krueger (2002) use clever methods to control for (some) "unobservables"
        - with repeated observations we can control for *some* time-invariant unobservables (*forthcoming*).
2. How do we know that the controls have entered the model in the correct way.
    - Can test for model specification w.r.t. included regressors

Functional Form Misspecification

Another way in which the model may be misspecified is that is excludes nonlinear functions of the **included variables**.

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + u_i$$

If MLR 4 holds, then if you were to add nonlinear functions the regressors, they should be irrelevant.

> Limitations
>
> This test cannot be used to test for other kinds of functional form misspecification such as whether the regressors should be included in log form. For this you need to look at nonnested tests (Wooldridge 2025, sec 9.1b).

RESET test

Ramsey's (1969) Regression Equation Specification Error Test (RESET) estimates the model,[7]

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \delta_1 \hat{y}_i^2 + \delta_2 \hat{y}_i^3 + \delta_3 \hat{y}_i^4 + u_i$$

where,

- $\hat{y}_i$ is the predicted value from the original equation.
- This regression has $n - k - 1 - 3 = n - k - 4$ degrees of freedom.

Null

You then test the null hypothesis,

$$H_0 : \delta_1 = \delta_2 = \delta_3 = 0$$

Options:

[7] Wooldridge (2025, 312) includes only the second and third polynomial of $\hat{y}$. However, Stata's default is to include the fourth-order polynomial.

- F-stat: $\overset{a}{\sim} F_{3,n-k-4}$ under H_0 (and MLR 1-5)
- LM-stat: $\overset{a}{\sim} \chi^2_3$ under H_0 (and MLR 1-5)
- Heteroskedastic-robust equivalents

In Stata/R

For example, we can use the RESET test to test for non-linearities in a simple Mincer equation.

Ramsey RESET (order 4)

```
library(lmtest)
model <- lm(lnwage ~ edu + exp, data=wage_data)
resettest(model, power = 2:4, type = "fitted")
```

```
    RESET test

data:  model
RESET = 36.675, df1 = 3, df2 = 4994, p-value < 2.2e-16
```

Example: Selective Education

Does this mean that, if you specify

$$y_i = \alpha + \beta p_i + \mathbf{x}_i\gamma + u_i$$

in which,

$$\beta = E[y_i|p_i = 1, \mathbf{x}_i] - E[y_i|p_i = 0, \mathbf{x}_i]$$

under MLR 4, then you have identified the *causal* impact of selective (private) education on earnings?

Answer: not clear.

- If β is a causal parameter, then yes.
- BUT what makes β a causal parameter?

Another control

Consider: what if I add another control z_i. The model will then give me,

$$\beta^{\ell} = E[y_i|p_i = 1, \mathbf{x}_i, z_i] - E[y_i|p_i = 0, \mathbf{x}_i, z_i]$$

where using the same OV-formula logic

$$\beta \neq \beta^{\ell}$$

Is **this** now the 'true' model/parameter?

Good controls?

And should anything that is correlated with y_i and p_i be included as a control?

- The OV formula tells us how the population parameter changes when we add additional regressors.
  - Correlations seem to be the important factor.
- So, should we add (in this example)?
  - Occupation of graduate job: should be correlated with y_i and p_i.
  - What about parental income?
  - And marital status?

The OV formula does not really answer these questions.

Summary

The OV formula provides a valuable tool for understanding:

- relationship between the short and long regression parameters
  - i.e., impact of adding additional regressors
  - gives us the *bias* in a model misspecified by omitted variables

BUT, doesn't tell us

- which covariates to add

- how they should appear in the model (interactions, non-linear terms, etc.)
- why the long model is 'true'

Modern: Conditional Independence

POF

Recall, we have a framework within which we have defined a causal effect:

$$\tau_i = y_i(1) - y_i(0)$$

- The problem is the ITE cannot be observed or estimated.
- We have to compare different groups of people, which introduces selection.

Selection

In Lecture 6 we derived this **important** decomposition:

$$\begin{aligned}
&E[y_i|w_i = 1] - E[y_i|w_i = 0] \\
=&E[y_i(1)|w_i = 1] - E[y_i(0)|w_i = 0] \\
=&\underbrace{E[y_i(1)|w_i = 1] - E[y_i(0)|w_i = 1]}_{} + \underbrace{E[y_i(0)|w_i = 1] - E[y_i(0)|w_i = 0]}_{} \\
=&\underbrace{E[y_i(1) - y_i(0)|w_i = 1]}_{\tau_{att}} + \underbrace{E[y_i(0)|w_i = 1] - E[y_i(0)|w_i = 0]}_{\text{selection}}
\end{aligned}$$

Unconfoundedness

We dealt with selection by exploiting the property of unconfounded assignment.[8]

> **Unconfounded assignment:** assignment into the treatment is unconfounded if

$$y_i(1), y_i(0) \perp w_i|\mathbf{x}_i$$

[8] Also referred to as (strong) ignorability (see Wooldridge 2025, 100).

Unconfoundedness

Most definitions of unconfoundedness include the condition on a finite set of observable covariates. You can think of the unconditional version (from Lecture 6) as a special case of this: where assignment is unconfounded independent of observables.

CIA

- Unconfoundedness is a property of assignment
  - *conditional* unconfoundedness may be the result of random assignment, stratified on $\mathbf{x}$, in a controlled experiment.
- In the absence of controlled assignment, unconfoundedness becomes an **assumption**
  - often referred to as the *Conditional Independence* **Assumption** (CIA)
  - *as if* treatment is randomly assigned, **conditional on $\mathbf{x_i}$**.

Selection on observables

CIA tells you,

$$E[y_i(0)|w_i = 1, \mathbf{x}_i] = E[y_i(0)|w_i = 0, \mathbf{x}_i] = E[y_i(0)|\mathbf{x}_i]$$

which means,

$$\underbrace{E[y_i(0)|w_i = 1, \mathbf{x}_i] - E[y_i(0)|w_i = 0, \mathbf{x}_i]}_{\text{Selection}(\mathbf{x}_i)} = 0$$

- Conditional $\mathbf{x}_i$, there is no selection.
  - selection on observables: YES
  - selection on unobservables: NO

In addition,

$$E[y_i(1)|w_i = 1, \mathbf{x}_i] = E[y_i(1)|w_i = 0, \mathbf{x}_i] = E[y_i(1)|\mathbf{x}_i]$$

Which means,

$$\underbrace{E[y_i(1) - y_i(0)|w_i = 1, \mathbf{x}_i]}_{\text{ATT}(\mathbf{x}_i)} = \underbrace{E[y_i(1) - y_i(0)|w_i = 0, \mathbf{x}_i]}_{\text{ATU}(\mathbf{x}_i)} = \underbrace{E[y_i(1) - y_i(0)|\mathbf{x}_i]}_{\text{ATE}(\mathbf{x}_i)}$$

These are conditional causal estimands:

- ATE for individuals with characteristics $\mathbf{x}_i$
- ATT for individuals with characteristics $\mathbf{x}_i$
- ATU for individuals with characteristics $\mathbf{x}_i$

Conditional comparison

$$\begin{aligned} & E[y_i|w_i = 1, \mathbf{x}_i] - E[y_i|w_i = 0, \mathbf{x}_i] \\ = & \underbrace{E[y_i(1) - y_i(0)|w_i = 1, \mathbf{x}_i]}_{\text{ATT}(\mathbf{x}_i)} + \underbrace{E[y_i(0)|w_i = 1, \mathbf{x}_i] - E[y_i(0)|w_i = 0, \mathbf{x}_i]}_{\text{Selection}(\mathbf{x}_i)} \end{aligned}$$

Under unconfoundedness (or CIA),

$$E[y_i|w_i = 1, \mathbf{x}_i] - E[y_i|w_i = 0, \mathbf{x}_i] = \underbrace{E[y_i(1) - y_i(0)|w_i = 1, \mathbf{x}_i]}_{\tau_{att}(\mathbf{x}_i)}$$

And,

$$\underbrace{E[y_i(1) - y_i(0)|w_i = 1, \mathbf{x}_i]}_{\tau_{att}(\mathbf{x}_i)} = \underbrace{E[y_i(1) - y_i(0)|\mathbf{x}_i]}_{\tau_{ate}(\mathbf{x}_i)}$$

BUT $\tau_{ate}(\mathbf{x}_i)$ (conditional) need NOT equal τ_{ate} (unconditional).

Matching

The CIA suggests that we need to **match** individuals with the same xs.

- upon matching, it is *as if* assignment was randomized
- the problem is the "dimensionality" of $\mathbf{x}_i$
  - same gender, ethnic background, parental income, number of siblings, etc.

Approaches to matching:

- covariate matching: make comparisons for every possible combination
- propensity score matching: match on $\rho(\mathbf{x}_i) = Pr(w_i = 1|\mathbf{x}_i)$ (the propensity score)

Regression as matching

When we wrote down the multivariate regression model in Lecture-1,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + u_i$$

with $x_1 \in \{0, 1\}$, we said

$$\beta_1 = E[y_i|x_{i1} = 1, x_{i2}, x_{i3}] - E[y_i|x_{i1} = 0, x_{i2}, x_{i3}]$$

under MLR 1-4.

Linear model

Suppose (for now) TEs are **homogeneous** (from Wooldridge 2025, 101):[9]

$$y_i = y_i(0) + \tau w_i$$

Then,

$$E[y_i|w_i, \mathbf{x}_i] = E[y_i(0)|w_i, \mathbf{x}_i] + \tau w_i = E[y_i(0)|\mathbf{x}_i] + \tau w_i$$

where second "=" follows from CIA.

[9]What is important here is just that $\tau_{ate}(\mathbf{x}_i) = \tau_{ate}$ (i.e. heterogeneity is independent of $\mathbf{x}_i$)

Let us assume then,

$$E[y_i(0)|\mathbf{x}_i] = \alpha + \mathbf{x}_i\gamma$$

i.e. linear regression function

Then,

$$E[y_i|w_i, \mathbf{x}_i] = \alpha + \tau w_i + \mathbf{x}_i\gamma$$

Then by the CEF theorem, we can write:

$$y_i = \alpha + \tau w_i + \mathbf{x}_i\gamma + u_i$$

where $E[u_i|w_i, \mathbf{x}_i] = 0$ (i.e. MLR 4)

- OLS is unbiased estimator for τ.

Assumptions:

Let us keep track of the assumptions we have just made:

A-1: CIA

A-2: Homogenous TEs: $\tau_i = \tau$

A-3: CEF of $y(0)$ is linear (in parameters): $E[y_i(0)|\mathbf{x}_i] = \alpha + \mathbf{x}_i\gamma$

Control variables

So, we have a linear regression model,

$$y_i = \alpha + \tau w_i + \mathbf{x}_i\gamma + u_i$$

- which identifies the causal impact of w on y
- by "controlling" for $\mathbf{x}$
  - the regression "matches" on x

Common Support

One of the key requirements of a matching estimator is "common support" (also referred to as overlapping). Formally, this means

$$0 < \rho(\mathbf{x}_i) < 1$$

That is, the conditional probability of treatment cannot be 0 or 1 (denoted above by the propensity score). From an experimental perspective, this is equivalent to saying at least one individual must be treated (untreated).
Without this assumption, you cannot match treated individuals with control individuals for every combinations of the x's. So, it is a crucial ingredient for matching estimators. Linear regression models do not impose this requirement. In fact, you could theoretically estimate the theoretical model with a complete absence of common support, leveraging the strength of the CEF (of $y(0)$) assumption. This assumption allows you to predict out the counterfactual for the treated group, even in the absence of common support.
This issue is discussed in Wooldridge (2025, 686, 691–92).

Good controls

So what xs should you control for?

- Evidently, we care about x's that determine selection (into treatment)
- As with RCTs, the xs **should not be an outcome of the "experiment"** (i.e. treatment)

  "Bad controls are variables that are themselves outocome variables in the notional experiment at hand. That is, bad controls might just as well be dependent variables too." (Angrist and Pischke 2009, 64)

> **Note**
>
> If you are unsure whether a variable could be a good control, ask yourself: could this be an outcome of the "treatment" variable (i.e. variable of interest).

Heterogeneity

We made a strong assumption: $\tau_i = \tau$.

- This seems unrealistic!
- Solution: Unrestricted Regression Adjustment (see Wooldridge 2025, sec. 7.6a) [10]

$$y_i = \alpha + \tau_{ate} w_i + w_i \cdot (\mathbf{x}_i - \bar{\mathbf{x}})\delta + \mathbf{x}_i\gamma + v_i$$

where,

- $w_i \cdot (\mathbf{x}_i - \bar{\mathbf{x}})$ is an interaction of w_i with each of the j demeaned controls.

$$w_i \times (x_{i1} - \bar{x}_1) + w_i \times (x_{i2} - \bar{x}_2) + \cdots + w_i \times (x_{ij} - \bar{x}_j)$$

Control

Can you "control" your way out of selection (and into causation)?

- Theoretically, YES
- Practically, in *observational studies*, it is often difficult to convince the reader that you observe *all* important variables for selection.
- Some policy settings can be a little "cleaner"

> **Policy example**
>
> Almus and Czarnitzki (2003) paper on *The effects of public R&D subsidies on firms' innovation activities: the case of Eastern Germany* exploits the fact that the government awarded grants based on a score that incorporated known criteria: size, past R&D, patents, sector, etc. While they do not know the exact scoring function, they do know the covariates upon which firms should be matched.

[10] See proof on p. 256 for more detail.

Example: Selective Education

If you assume *A-1,2,3*, then the model

$$y_i = \alpha + \tau_{ate} p_i + \mathbf{x}_i \gamma + u_i$$

DOES, indeed, identify the causal impact of selective education.

Traditional Modern

We can "marry" these two approaches.

> **Note**
>
> For this section, we will do so for using the notation in the running example.

"Marriage"

We will show,

1. For a discrete variable (of interest), the OV formula tells you about selection into "treatment".

$$\text{Traditional} \Rightarrow \text{Modern}$$

2. When comparing two groups, the selection term is given by the OVB term (under **A-1,2,3**)

$$\text{Modern} \Rightarrow \text{Traditional}$$

This "marriage" is not always so straightforward. The discrete nature of the "treatment" (private education) considered here simplifies things.

- As always, heterogeneous TEs complicate things.

From the Traditional

In the short (simple) model

$$y_i = \alpha^s + \beta^s p_i + u_i$$

we know that,

$$\beta^s = E[y_i|p_i = 1] - E[y_i|p_i = 0]$$

Applying POF, this is

$$\beta^s = \underbrace{E[y_i(1) - y_i(0)|p_i = 1]}_{\tau_{att}} + \underbrace{E[y_i(0)|p_i = 1] - E[y_i(0)|p_i = 0]}_{Selection}$$

Substituting in the longer model (here simplified with a single x),

$$y_i = \alpha + \beta p_i + \gamma x_i + v_i$$

you get,

$$\begin{aligned}
\beta^s =& E[\alpha + \beta p_i + \gamma x_i + v_i|p_i = 1] - E[\alpha + \beta p_i + \gamma x_i + v_i|p_i = 0] \\
=& \underbrace{\alpha - \alpha}_{=0} + \beta + \gamma[E[x_i|p_i = 1] - E[x_i|p_i = 0]] + \underbrace{[E[v_i|p_i = 1] - E[v_i|p_i = 0]]}_{=0} \\
=& \beta + \gamma[E[x_i|p_i = 1] - E[x_i|p_i = 0]]
\end{aligned}$$

The term,

$$E[x_i|p_i = 1] - E[x_i|p_i = 0] = \frac{Cov(x_i, p_i)}{Var(p_i)}$$

is the slope parameter from a regression of x on P.

- This was δ_1 in the OV formula

This can be rewritten as,

$$\frac{Cov(x_i, p_i)}{Var(p_i)} = \underbrace{\frac{Cov(x_i, p_i)}{Var(x_i)}}_{\theta_1} \times \frac{Var(x_i)}{Var(p_i)}$$

where θ_1 is from,

$$p_i = \theta_0 + \theta_1 x_i + \epsilon_i$$

- A linear probability model of selection.
- If $\theta_1 = 0$: **selection** is unrelated to x.

The OVB formula in this setting is,

$$OVB = \beta^s - \beta = \gamma\delta_1 = \gamma\theta_1 \frac{Var(x_i)}{Var(p_i)}$$

- If selection is unrelated to x, then OVB = 0
- Fits with the idea of selection on observables, alone.

From the Modern

Under **A-1,2,3**, outlined earlier,

$$\begin{aligned}\beta^s =& E[y_i|p_i = 1] + E[y_i|p_i = 0] \\ =& \underbrace{E[y_i(1) - y_i(0)|p_i = 1]}_{ATT} + \underbrace{E[y_i(0)|p_i = 1] - E[y_i(0)|p_i = 0]}_{\text{Selection}}\end{aligned}$$

Under *A-3*,

$$y_i(0) = \alpha + \gamma x_i + \varepsilon_i$$

Where, given **A-1** and **A-3**:

$$E[\varepsilon_i | p_i, x_i] = E[\varepsilon_i | x_i] = 0$$

The selection term is therefore,

$$\begin{aligned}
&E[y_i(0)|p_i = 1] - E[y_i(0)|p_i = 0] \\
=&E[\alpha + \gamma x_i + \varepsilon_i|p_i = 1] - E[\alpha + \gamma x_i + \varepsilon_i|p_i = 0] \\
=&\underbrace{\alpha - \alpha}_{=0} + \gamma \underbrace{[E[x_i|p_i = 1] - E[x_i|p_i = 0]]}_{\delta_1} + \underbrace{[E[\varepsilon_i|p_i = 1] - E[\varepsilon_i|p_i = 0]]}_{=0 \text{ by CIA}} \\
=&\gamma\delta_1
\end{aligned}$$

Since $\beta = \tau_{att} = \tau_{ate}$ (under **A-1** and **A-2**)

$$\beta^s = \tau_{att} + \text{Selection} = \beta + \gamma\delta_1$$

which is the OV formula.

- the multivariate regression model is explicitly modelling ("controlling for") selection, *under these assumptions*.

Example: Selective Education

Identification strategy

Novel identification strategy to control for factors that may normally be unobserved (i.e. omitted).

- Selection into selective universities is based on a number of factors; some explicit (school, grades, sporting achievements), some implicit (social class, familial connections).
- Ultimately, it *might be* reasonable to assume that if two individuals received offers from the same set of schools, then they had "comparable/similar" profiles.
  - Of course, you could pick holes in this assumption.

Consider the following models:

- Unconditional

$$y_i = \alpha^s + \beta^s p_i + u_i$$

- Conditional on observables: SAT and individual characteristics/demographics

$$y_i = \alpha + \beta p_i + \gamma_1 SAT_i + \mathbf{x}_i\gamma_2 + v_i$$

- Conditional on "unobservables": PLUS application group dummies (fixed effects)

$$y_i = \alpha^\ell + \beta^\ell p_i + \gamma_1^\ell SAT_i + \mathbf{x}_i\gamma_2^\ell + grp_i\psi^\ell + \varepsilon_i$$

Dale & Krueger

The take home message from Dale and Krueger (2002):

- in observational studies, where you know little about selection, it can be quite unrealistic to "control" for all important confounding factors.
- this need not hold in policy settings - e.g. Almus and Czarnitzki (2003) - where you know the determinants of the "treatment".
- we need to consider other procedures to deal with this problem of selection: for example, instrumental variables and repeated observation.

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